



Antonio Rizza

Postdoctoral Researcher

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 antoniorizza95

Languages

Italian, English, Spanish, French

- ## Areas of Expertise
- Spacecraft GNC
 - Astrodynamics and Formation Flying
 - Modeling and Simulation of Space Systems
 - Dynamical Systems Theory
 - Optimal Control Theory
 - Non-linear State Estimation

- ## Computer Skills
- MATLAB/Simulink
 - C++, Python (VSCode)
 - NASA tools (SPICE, GMAT)
 - CAD tools (SolidWorks, Solid Edge, Inventor)
 - Editors/Styles (LaTeX, Office, VSCode)
 - OS (iOS, Linux, Windows)

Headline

Spacecraft Guidance Navigation and Control (GNC) Engineer. Expert in modeling, simulation and testing of dynamical Space Systems, spacecraft autonomy, precise formation flying and astrodynamics. Seven years of experience in research on autonomous space robotics probes, with leading GNC roles for both Low Earth Orbit (LEO) and deep-space missions sponsored by national and international agencies across Europe and the US. Currently focused at Stanford on enabling and flight testing ground braking GNC technologies for distributed space systems that can be pivotal in the current landscape of Space exploration and exploitation. GNC Engineer for the [Milani](#), [LUMIO](#), and [STARI](#) mission.

Research Experience

01/2025 - ongoing	Postdoctoral Scholar	Stanford University - SLAB team
	Enabling safe, robust and high precision GNC solutions for the next generation of distributed space systems. Lead GNC engineer for the STARI mission.	
11/2024 - 12/2024	Research Fellow (Senior)	Politecnico di Milano - DART team
	Developing guidance and control solution for proximity operations around small bodies. Supporting the development of a robotic facility for validation and verification of GNC algorithm.	
01/2023 - 12/2023	Visiting Research Scholar	Stanford University - SLAB team
	Goal-oriented trajectory optimization about asteroids using sequential convex programming. Hardware in the loop validation of Active SLAM approaches using the robotic testbed for rendezvous and optical navigation at SLAB.	
11/2021 - 11/2024	PhD cum laude in Aerospace Engineering	Politecnico di Milano - DART team
	Goal-oriented guidance under uncertainties for small bodies exploration. GNC engineer for the Milani and LUMIO missions.	
11/2020 - 10/2021	Research Fellow	Politecnico di Milano - DART team
	Development of high-fidelity six-degrees-of-freedom simulator for autonomous deep-space missions.	
10/2019 - 06/2020	M.Sc. Thesis	VKI, Belgium
	Simulation and testing of gas-surface interaction phenomena in atmospheric re-entry conditions for Design for Demise analysis.	
08/2017 - 09/2017	Research Internship	VKI, Belgium
	Software developer for the VKI Expandable Stability and Transition Analysis toolkit to predict boundary layer transition in high speed flow.	

Education

09/2017 - 06/2020	M.Sc. cum laude in Space Engineering	Politecnico di Milano
05/2019	ESA Academy Space Debris Training Course	ESEC GALAXIA, Belgium
10/2014 - 09/2017	Bachelor's in Aerospace Engineering	Politecnico di Milano

Research projects and Missions

01/2026-ongoing	Flight Demonstration of Autonomous Navigation in GEO Collaboration with Blue Origin to demonstrate in GEO for the first time autonomous navigation capabilities based on angles only navigation and CNN-based relative pose estimation to known targets.
01/2025-ongoing	FALCON - Research Co-lead Fast Autonomous Lost-in-space Catalog-based Optical Navigation (FALCON) is a new alternative space-based Positioning, Navigation and Timing (PNT) system for responsive Space Situational Awareness (SSA) of on-orbit assets.
01/2025-ongoing	STARI - Orbit and GNC Engineer - Phase A to E The STarlight Acquisition and Reflection toward Interferometry (STARI) mission is a technology demonstration designed to advance key technologies for future space-based interferometry. The mission makes use of two 8U CubeSats flying in close formation, effectively simulating one arm of a space interferometer.
09/2022-12/2024	LUMIO - AOCS Engineer - Phase B The Lunar Meteoroid Impact Observer (LUMIO) is a CubeSat mission to a quasi-halo orbit at the Earth-Moon L2 Lagrangian point designed to complement ground based observations and characterize the lunar meteoroid environment.
11/2020-06/2022	Milani - GNC Engineer - Phase A to D The (Milani) CubeSat is part of ESA's Hera mission, the European contribution to the ESA-NASA collaboration AIDA (Asteroid Impact & Deflection Assessment) around the binary asteroid 65803 Didymos.
11/2020-06/2022	PRESS Proposal for Rams Enhancing in Small Satellites (PRESS) is a project aimed at improving Reliability, Availability, Maintainability and Safety (RAMS) for CubeSats reducing their currently high failure rate.

Teaching and tutoring

Grading and tutoring graduate students for the course AA279D *Control of Distributed Space Systems* at Stanford University.

Teaching assistant for the graduate course in *Spacecraft Attitude Dynamics and Control* at Politecnico di Milano for two years.

Co-advisor of PhD, master and undergraduate students at the Space Rendezvous Laboratory (SLAB) at Stanford University.

Co-advisor of tens of M.Sc. students during their final thesis work at the Deep-space Astrodynamics Research and Technology (DART) group at Politecnico di Milano.

Publication records

<https://scholar.google.com/citations?user=JqvWV-AAAAAJ&hl=it>